Linux Fundamentals

Program Code: XE103

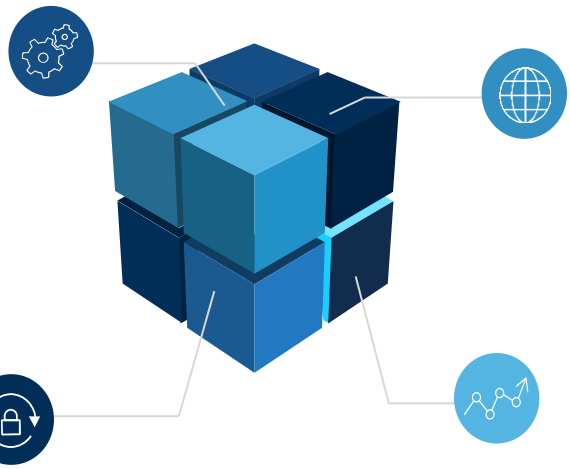
OPERATION ORDER

Operation **Info Extractor** is a straight-to-the-point mission with one clear goal: "Gather System Info." This task involves creating a smart script to dig into the system and pull out key details. You'll need to find out network info like IP and MAC addresses, check how the system is running by looking at CPU and memory use, and figure out which files and folders are the biggest. Your mission is to do this neatly and efficiently, giving us a clear picture of what's going on inside the system.



1. VISION

To empower users with optimization of their digital environments, providing tools for enhanced understanding and security. We aim to simplify complex diagnostics into actionable ones.



2. MISSION

Simplify complex system diagnostics, ensuring enhanced security and performance.

3. STRATEGY

Create a script to display the system status.

4. OBJECTIVES

- Create intuitive tools, making them accessible to users of all skill levels.
- Educate users on the system's health and security.
- Continuously improve and update our offerings in response to evolving technology and user feedback.

Project Structure

System Diagnostics

1. Identify the system's public IP.
2. Identify the private IP address assigned to the system's network interface.
3. Display the MAC address (masking sensitive portions for security).
4. Display the percentage of CPU usage for the top 5 processes.
5. Display memory usage statistics: total and available memory.
6. List active system services with their status.
7. Locate the Top 10 Largest Files in /home.

General

Tools and Commands: **curl**, **ifconfig/ip addr**, **top**, **ps**, **du**, **find**, standard Bash scripting utilities.

Submitting Your Project

To ensure a successful submission of your project, please adhere to the instructions provided below. Following these guidelines will facilitate a smooth evaluation process and allow you to receive feedback on your work promptly.

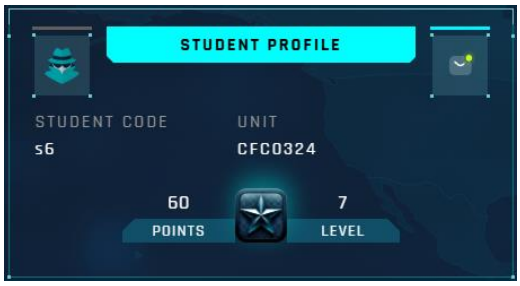
1. What to Submit

Only .sh and .pdf files will be accepted. **No other file types** (e.g., .docx, .txt) will be accepted. Ensure your files are in the correct format before submission.

You must submit both the script (.sh) and the corresponding PDF. The PDF should prove that you completed all the tasks and that the project works as expected. If you miss the script file or the PDF, you will be asked to resend your submission.

2. File Naming

Ensure your project files follow the correct naming convention:
UNIT.STUDENT.PROGRAM



For example, if your unit is CFC0324, your student ID is s6, and your program code is xe103, the correct filenames should be:

CFC0324.s6.xe103.sh
CFC0324.s6.xe103.pdf

3. Submission Process

Email Address: Forward your project materials to **projects@thinkcybergroup.com** and to your trainer.

Email Subject: Format your email subject as follows: **Project: INFO EXTRACTOR <Student Name:Unit Name>**. Make sure to replace **<Student Name>** with your actual name and **<Unit Name>** with the precise name of the unit as shown in the simulator.

4. **Unique Work**

   Your work **must** be unique. If the system detects that your project has been copied from another student, **both** the student who copied the work **and** the student who allowed their work to be copied will be **banned**. After being banned, the server will no longer check your future submissions.

   Important: You are encouraged to help each other with ideas and techniques, but each student **must submit their own, original project**. If you submit a project that is flagged as identical or too similar to another student's, both of you will be banned from submitting future work to the system.

5. **Use of AI Tools**

   You are allowed to use tools like ChatGPT or other AI bots for assistance, but your program must work and be clear. If your AI code is simply copied, doesn't work, or is unclear, you will be banned from future submissions.

Response Time: After submitting, you can track the status of your submission (pass or fail) through the ***Progress Tracker*** inside your student account. Allow up to one week for your project to be reviewed.

How to Showcase Your Cybersecurity Projects on LinkedIn

1. **Write a Compelling Post**:
   When you share your project on LinkedIn, make sure to write a post that highlights the importance of the project and your key takeaways from it. Use engaging language and explain:
   - **The challenge** you solved.
   - **Tools** and technologies you used (mention relevant tools like Python, Shell Scripting, Scapy, etc.).
   - **Skills** you developed (problem-solving, incident response, penetration testing, etc.).
   - **Real-world relevance** of your project. Explain how it can be applied to secure networks, detect vulnerabilities, or analyze threats in a professional setting.

Example:

"Excited to share my latest project in Network Security (ZX305) where I built a script to detect network anomalies. This project helped me understand intrusion detection and how to safeguard systems against real-world threats. Using Python and Shell scripting, I developed a tool that can assist security teams in monitoring and reacting to suspicious network activities."

2. **Attach Your PDF**:
   Upload the PDF file of your project as an attachment to your post. This will allow recruiters, potential employers, or other professionals in the cybersecurity field to directly review your work.
   - Ensure your **PDF is professional**, free of errors, and clearly explains each section of the project.
   - **Highlight results**: If you were able to detect or simulate cyber threats, mention how your tool worked and the outcomes of your project.

3. **Record a Video Demo**:
   Videos are powerful on LinkedIn. Record a quick demo of your project where you:
   - Explain the problem.
   - Walk through your code.
   - Show your script in action (for example, showing how your network security tool detects a threat).

Example Intro:

"Here's a quick demo of my project where I developed a Python-based tool to monitor network traffic and detect potential cyber threats. Watch how it scans real-time traffic and flags suspicious activity."

4. **Mention the Certificate**:
   Once you complete the project, your simulator generates an official **ThinkCyber certificate**. Share this alongside your post to demonstrate your verified completion of the project.

Example:

"Proud to have completed my Network Security (ZX305) program with ThinkCyber! This certificate is proof of the skills I've acquired in identifying and mitigating cybersecurity threats."



Make sure to upload the certificate as an image. Use relevant hashtags like **#cybersecurity**, **#networksecurity**, and **#Python** to increase your visibility.

5. **Encourage Networking**:
   Mention that you are open to connecting with cybersecurity professionals and recruiters who might find your skills and projects relevant to their needs.